

House-price Prediction Based on OLS Linear Regression and Random Forest

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ABSTRACT

Based on a simple dataset with 20 features of houses, this paper builds two forecasting models to predict house prices. It primarily takes a series of data processes, including missing value process, outlier removal, new variables creation, and correlation analysis. Then this paper conducts the first theoretical model using OLS method. Next, it uses secondary cross-validation to optimize the parameters of random forest and gets a respectively realistic model. Finally based on the results, this paper analyzes and compares the advantages and disadvantages of the two models.

CCS CONCEPTS

• **Applied computing**; • **Operations research**; • **Consumer products**;

KEYWORDS

OLS linear regression, random forest, house-price forecasting

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1 INTRODUCTION

In the house-price market, the most growing topic in research is ‘forecasting’ as its importance and popularity among the masses and also companies of different scales due to financial benefits and its low risk. The real estate industry accounts for a large part of GPA, and it can directly influence citizens’ consumption level as well as the overall domestic economy. Thus, accurate house-price forecasting not only benefits investors and consumers but also is a foundation of steady economic development. Attributing to the weakness of traditional methods, such as subjectivity and empiricism, some scholars proposed the Hedonic method to improve the accuracy. The Hedonic method, combined with OLS regression method, and random forest has been widely applied in house price predicting [1].

Based on the Hedonic model, Thibodeau [2], Selim [3] studied the impact of environmental characteristics on house prices, built

multiple linear regression to predict house prices. They found that the house area, age, proximity to transportation, school, and scenic view are the most important environmental factors on house price. Bourassa, Cantoni, & Hoesli [4] compared OLS linear regression and other three models to explore the characteristic of OLS in the relationship between spatial pattern and house prices. They presented that OLS emphasize neighborhood factors but ignore spatial structure. Anderson & West [5], Crompton [6], Dehring & Dunse [7] concluded house sales price and a variety of open space, such as urban parks, land in conservation easements, agriculture farmlands, golf site, and other greenbelts have positive relationships by hedonic methods.

Antipov, Fan et al. [8] used the tree decision to evaluate house characteristics and to predict house price. They concluded random forest is stable in controlling outliers and effective in missing values as well as categorical variables. Hong [9] maintained that traditional hedonic pricing method is unstable and inaccurate, but random forest is competent because it can capture the complexity and non-linear relationship in practical operation. Yoo [10] applied machine learning regression methods to house prices prediction to select feature variables and compared the results using machine learning with traditional OLS method. He found that random forest is useful in selecting importance of variables for the hedonic price equation because there is an in-depth hypothesis of variables showing customers’ preferences by random forest method. Ceh [11] used GIS explanatory dataset, such as structural and age features of the houses, and neighborhood information to process random forest model. He also measured the performance using R², MAPE, and COD (coefficient of dispersion), revealing that machine learning method on house price prediction is prospective.

OLS linear regression and random tree model are some of the most popular forecasting models. The former is easier to understand and focuses on explaining the correlatives among variables. The latter is bagging learning and needs a number of arithmetic operations. It can deal with non-linear and overfitting problems. Therefore, this research aims to use the two methods in house price forecasting and compare the difference between them.

2 DATA PREPROCESSING

2.1 Data Access and Description

The house price dataset in USA, comes from Kaggle, containing 21613 samples with 20 features, which is suitable for this study both in data size and features. This dataset includes architectural features of houses (e.g. number of waterfronts, time of yr_built, number of bedrooms), location of residential houses (e.g. latitude and longitude), and other features (e.g. price and grade). As there is no null value in the dataset, we skip processing the missing value.

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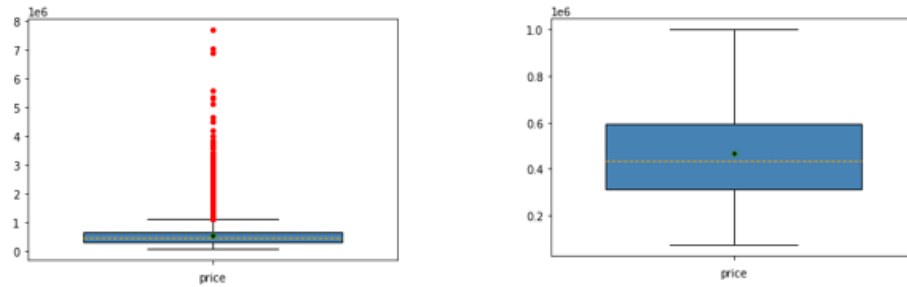
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Table 1: Category of 20 Feature Variables

Variable Type	Architectural Features	Geographic Features	Others
Categorical Variable	waterfront, yr_built, yr_renovated,	zipcode, lat, long	id, date
Numerical Variable	bedrooms, bathrooms, sqft_living, sqft_lot, floors, sqft_living15, sqft_lot15		price, view, condition, grade,

Table 2: The Description of Variables

Variable	Description
sqft_living	Square footage of the apartment interior living space
grade	An index from 1 to 13, where 1-3 falls short of building construction and design, 7 has an average level of construction and design, and 11-13 has a high level of construction and design
sqft_above	Square footage of the apartment interior housing space that is above the ground level
sqft_living15	Square footage of the apartment interior living space for the nearest 15 neighbors
date	The date when the house was sold
yr_built	The year when the house was built

**Figure 1: Boxplot of Price Before and After Outliers Removal**

2.2 Outlier Removal

Boxplot is an effective method to detect outliers that are defined as the values smaller than $QL - 1.5IQR$ or larger than $QL + 1.5IQR$ (QL is lower quartile; QU is upper quartile). Draw a boxplot of price with the outliers highlighted in red, and then remove the outliers.

2.3 New Variable Creation and Variable Transformation

Because there is no variable indicating house age and whether the house is renovated, we create a new variable "built_year", expressing the house age, which is the year sold minus the year built. And create new variable "yr_renovated", expressed by 0 and 1 (0 represents no renovation).

Detecting the normal distribution of data, and there are some variables that were significantly skewed to the left or right, which would lead to heteroscedasticity in the regression analysis. For example, "sqft_above" needs applying logarithmic transformation due to its skewness 1.068. As a result, the skewness declines to 0.1736, meeting the requirement of normal distribution. We apply the logarithmic transformation to other variables whose skewness is greater than 1.

In this housing price predicting dataset, the units of each variable are different. To eliminate the adverse effect brought by the dimension of variables, they are all standardized.

2.4 Correlation Analysis

Due to a large number of variables, the correlation chart was made for variables with price correlations greater than 0.3. It can be seen that price has the strongest correlation with the score and house area, and also has a correlation with latitude and bathrooms.

3 METHODS

3.1 OLS Linear Regression

3.1.1 Theory of OLS Linear Regression. Linear regression model assumes regression function $E(Y|X)$ on the input variable $X_1, X_2, \dots, X_p$ linear, that is for input vector $XT = (X_1, X_2, \dots, X_p)$ and the output variables of y necessary to predict. The following form is:

$$f(X) = \beta_0 + \sum_{j=1}^p X_j \beta_j \quad (1)$$

In the formula β_j is an unknown parameter and needs optimizing. Variable X_j is a feature variable. It can come from:

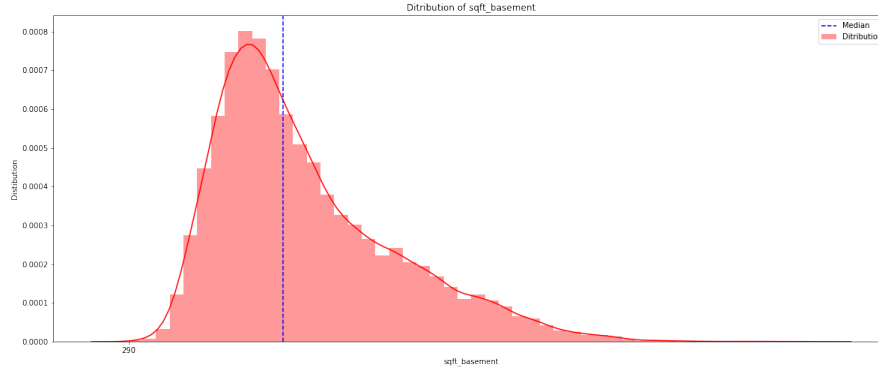


Figure 2: Distribution of Sqft_basement

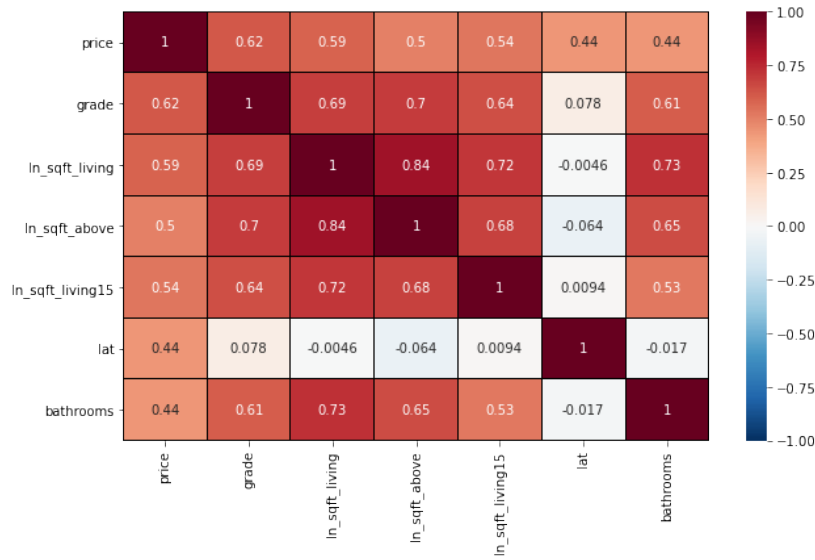


Figure 3: Correlation of Variables Greater than 0.3

- (1) Quantitative input variables.
- (2) Transform quantitative input variables, such as logarithmic transformation square root transformation, etc.
- (3) The polynomial generated by the expansion of the basis function, such as $X_2 = X_1$, $X_3 = X_1$
- (4) Sorting or dummy encoding of categorical variables. For example, X_j is a qualitative input variable with five levels, the five levels can be transformed into 1, 2, ..., 5. Or one hot coding, which constitutes several 0 and one 1.
- (5) Cross effects of variables, such as $X_3 = X_1 \cdot X_2$.

$$\begin{aligned}
 L(Y, f(X)) &= \sum_{i=1}^n (y_i - f(x_i))^2 \\
 &= \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)^2 \quad (2) \\
 &= (y - X\beta)^T (y - X\beta)
 \end{aligned}$$

This research sets the house price as the dependent variable Y and the variables after data processing as the independent variables X_j . The Sklearn function in Python is used to divide the data into 80% training set and 20% test set. Then use the least square method to give an optimal solution of the parameter $\beta = (\beta_0, \beta_1, \beta_2, \dots, \beta_j)$

Because X is full column rank, strives for the partial derivatives on loss function $L(Y, f(X))$, and make the first order differential equals to 0. The final solution is:

$$\tilde{\beta} = (X^T X)^{-1} X^T y \quad (3)$$

Then input feature vector x_0 , and predicted value is given by $\hat{f}(x_0) = (1 : x_0)^T \tilde{\beta}$

When the number of feature variables is large, the linear regression prediction model often has small deviation and large variance, namely, overfitting. In order to improve the accuracy of the model, a sacrifice will be made to shrink some feature variable coefficient that is not very explanatory to 0. This study chooses lasso regression,

Table 3: Outcome of OLS Linear Regression

Dep. Variable:	price	R-squared:	0.687
Model:	OLS	Adj. R-squared:	0.687
Method:	Least Squares	F-statistic:	1770
No. Observations:	16118	Prob (F-statistic):	0
Df Residuals:	16097	Log-Likelihood:	-13460
Df Model:	20	AIC:	2.70E+04
Covariance Type:	nonrobust	BIC:	2.71E+04

Table 4: Sort of Feature Importance

	coef	std err	t	P> t	[0.025	0.975]
if_renovated	-4.163	0.542	-7.680	0.000	-5.225	-3.101
built_year	3.195	0.442	7.230	0.000	2.329	4.061
grade	0.373	0.007	51.592	0.000	0.359	0.387
lat	0.372	0.005	79.101	0.000	0.363	0.381
ln_sqft_living15	0.168	0.007	23.668	0.000	0.154	0.181

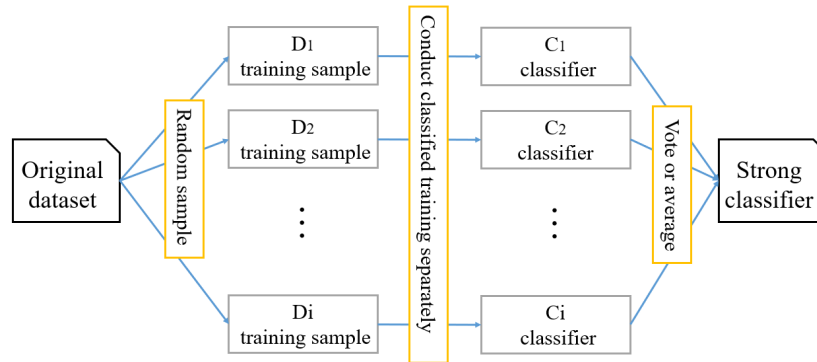


Figure 4: Bagging Structure

which is adding a penalty term to loss function, to avoid overfitting phenomenon.

$$L(Y, f(X)) = \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \sum_{j=1}^p |\omega_j| \quad (4)$$

3.1.2 Results of OLS Linear Regression. The training set was modeled and the results were shown in the following table. The significance level is passed. R^2 was 0.687 so the fitting effect was impractical, and the accuracy verified by the testing set was only 0.682.

The important features are recognized. The result presents consumers emphasize more on house condition and grade than spiral characteristics.

3.2 Random Forest

3.2.1 Theory of Random Forest Regression. Random forest is an Ensemble Learning algorithm, belonging to bagging. By combining several weak classifiers, the final results are voted or averaged, so

that the results of the overall model have high accuracy and generalization performance. Its good performance is mainly attributed to "randomness" and "forest", one making it resistant to overfitting and one making it more accurate.

Except for bagging algorithm, random forest applies a unique approach to node splitting. In traditional classification and regression trees, node splitting is usually conducted using all the predictors, whereas a random subset of the predictors is used to select on each node in random forest node splitting. When a tree is constructed, nearly one third of the observations are hardly applied to any individual trees, which is called out-of-bag (OOB) due to the hold-out cases. In the OOB approach, the errors excluded from the data generated by each regression tree are used to give random forest a notice of the relative advantages and correlation of the tree [12].

3.2.2 Secondary Cross Validation. In machine learning, generalization error is used to measure the accuracy of the prediction model. When the random forest model is complex, overfitting may appear, leading to large generalization error. In contrast, when the model is simple, underfitting will come out, with large generation error.

Table 5: Value Range of Secondary Cross Validation Parameters

Parameter	First Time	Second Time	Final
n_estimators	[1,200]	[167,176]	170
min_samples_split	[2, 5, 10]	[3, 9]	5
min_samples_leaf	[1, 2, 4]	[2, 3]	2
max_features	['auto','sprt']	['auto']	'auto'
max_depth	[10,100] or [None]	[76,84]	84
bootstrap	[True, False]	[True]	TRUE

Table 6: Sort of Feature Importance

Variable	Importance
lat	0.44
ln_sqft_living	0.32
grade	0.06
long	0.05
ln_sqft_living15	0.03
ln_sqft_lot	0.02
ln_sqft_lot15	0.02
bathrooms	0.01
view	0.01

Therefore, parameters such as max_depth and min_samples_leaf will be adjusted to reduce the depth, branches and complexity of the tree.

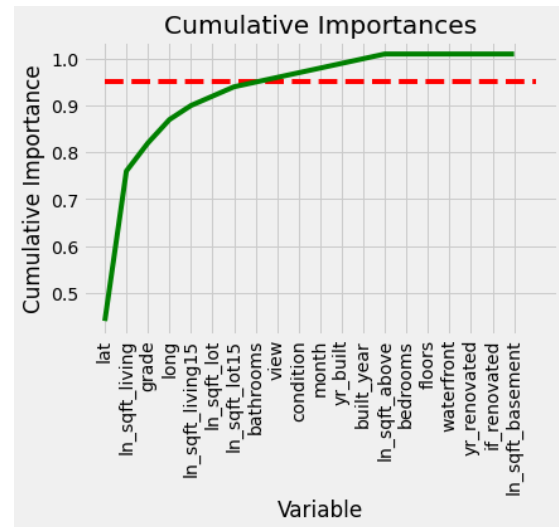
An optimal combination of parameters will significantly prompt the performance of a random forest model. This study uses 17290 out of 21613 samples as training set and uses the remaining 4323 samples to test the model. Training and testing sets are selected randomly according to the municipality to which each residential property belongs. In the course of cross validation, 80% of original training data are selected to train model and 20% of training sets are selected to validate the model. For the reason that there are various parameters with a large range to test in this model, secondary cross-validation is used to select the best combination of parameters.

3.2.3 Results of Random Forest. Use the optimal parameter to establish the random forest model, and get the final result, the accuracy is 0.8589, which precedes OLS model.

The figure below is the importance order of feature variables according to the random forest algorithm. The dotted red line represents the cumulative importance is 0.95. Four of the top five variables are related with geographical characteristics (latitude, sqft_living, longitude, and sqft_living15).

4 CONCLUSION

In terms of important feature ranking, although there are similar features, OLS ranks house age and renovation first, while decision tree focuses on spatial and geographical features. In terms of price prediction, decision tree fitting effect is obviously more practical

**Figure 5: Sort of Variables Importance using Random Forest**

than OLS in the face of complex and nonlinear data. Therefore, in these two conditions, one is a large number of observations in this dataset, the other is complex and has noise samples, we recommend to choose random forest.

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